QUASI-COMPACT SUBSETS OF A HAUSDORFF SPACE ARE CLOSED, AND DISJOINT QUASI-COMPACT SUBSETS CAN BE SEPARATED BY OPEN SETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of quasi-compactness for subsets with the induced topology, and the Hausdorff separation axiom. Set up in the author's sections on Hausdorff spaces and on quasi-compact spaces, before the counted statement. Fixes the hypothesis on the space and the two conclusions about quasi-compact subsets.

Lemma 0.1. *Let X be a Hausdorff topological space.*

(1) *If $E \subset X$ is quasi-compact, then it is closed.*
(2) *If $E_1, E_2 \subset X$ are disjoint quasi-compact subsets then there exists opens $E_i \subset U_i$ with $U_1 \cap U_2 = \emptyset$.*

Proof. Proof of (1). Let $x \in X$, $x \notin E$. For every $e \in E$ we can find disjoint opens V_e and U_e with $e \in V_e$ and $x \in U_e$. Since $E \subset \bigcup V_e$ we can find finitely many $e_1, \ldots, e_n$ such that $E \subset V_{e_1} \cup \ldots \cup V_{e_n}$. Then $U = U_{e_1} \cap \ldots \cap U_{e_n}$ is an open neighbourhood of x which avoids $V_{e_1} \cup \ldots \cup V_{e_n}$. In particular it avoids E. Thus E is closed.

Proof of (2). In the proof of (1) we have seen that given $x \in E_1$ we can find an open neighbourhood $x \in U_x$ and an open $E_2 \subset V_x$ such that $U_x \cap V_x = \emptyset$. Because E_1 is quasi-compact we can find a finite number $x_i \in E_1$ such that $E_1 \subset U = U_{x_1} \cup \ldots \cup U_{x_n}$. We take $V = V_{x_1} \cap \ldots \cap V_{x_n}$ to finish the proof. □

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